VISIBU · IMPROVEMENTS · NOVEMBER 2026

From 5 simple words, to “and and and”, to a sign interpreter that learns.

WHERE IT STARTED

“5 simple words”

Hand-written rules that could only recognise five basic signs.

→

FIRST MODEL

~~“and and and and”~~

A real model at last, but one guess, often nonsense, that blocked the screen.

→

NOW

“Hi, how are you doing?”

Two trained models, three suggestions, your own taught signs, and nothing sent until you say so.

This is everything I changed to take Visibu from a fragile prototype to a working, on-device ASL-to-English demo: new models, new data, cloud training, a rebuilt review experience, and a feature that lets people teach Visibu their own signs.

`2 trained models` `~125,000 clips of training data` `250 everyday signs`

`29,547 sentences` `+72% sentence score` `54 + 11 automated tests`

`0 video leaves the device`

First, the honest part: this is a demo

Visibu will sometimes misread a sign or write the wrong sentence. ASL is rich, personal and regional, and open training data is scarce. So instead of pretending to be perfect, Visibu is built to **learn**: every message is checked by the signer before it is shown, and with **My signs** anyone can show Visibu a sign and type what it means.

01

THE BRAIN TRANSPLANT

I replaced the guesswork with a real translation pipeline

Visibu used to know just 5 simple words, from hand-written rules. I swapped those rules for a trained model that reads continuous signing, the way a person signs.

- **Camera to landmarks:** MediaPipe tracks 553 points on the hands, face and body, 15 times a second. Only these numbers are used, never the video.
- **One feature recipe, two languages:** the same 786-number description of each frame is computed in Python for training and in the browser for the app, with a test that fails if they ever disagree.
- **A Transformer translator:** a 6-layer sign encoder, a helper head that spots words, and a sentence decoder.
- **Runs in the browser:** exported to ONNX and run with ONNX Runtime Web inside a background worker, so the page never freezes. Every model file is checked against its fingerprint (sha256) before use, and the security policy was kept intact.

02

KILLING “AND AND AND”

One bad guess became three good options

The first trained model blurted out its single most likely word over and over: “and and and and”. Now it searches properly.

- **Beam search:** keeps four candidate sentences alive at once, blocks repeated phrases, and never returns an empty answer. Python and browser versions give identical results, word for word.
- **Three suggestions per translation,** each with its score, so the signer picks the closest one instead of fighting one guess.
- **Faster in the browser:** the decoder now handles all candidates in one pass instead of one at a time.

03

THE DATA RESCUE

When the downloads broke, I went and got the data anyway

How2Sign’s 31 GB video archive hit Google Drive limits and arrived as broken files over a slow connection. Instead of stopping there:

- **Rescued 4,488 clips** out of the broken zip files and turned them into landmarks for a first sentence model.
- **Found a second dataset:** Google’s Kaggle “Isolated Sign Language Recognition” data, with 94,477 clips of 250 everyday signs from 21 people, already as landmarks. Its licence allows commercial use with credit.
- **Fixed a hidden mismatch:** that data never says what shape its videos were, so I measured it from face proportions (cheek-to-cheek vs forehead-to-chin) against How2Sign and got portrait 3:4.
- **Kept the tests fair:** people used for testing never appear in training, so every score is on signers the model has never seen.

04

INTO THE CLOUD

I moved the heavy training to free cloud GPUs

Training on the laptop lagged the machine for hours. So the heavy lifting moved to Kaggle's GPUs, and only the finished model comes back.

- **Sign pre-training on Kaggle:** a private notebook built from the exact tested code. About 4 hours on GPU, with no download on our side.
- **The full How2Sign, without the videos:** I found the whole corpus already turned into landmarks ("How2Sign Holistic") and checked it against our own extraction: the features matched within about 1%. Then the cloud trained on all 29,547 sentences in 2.4 hours.
- **Kind to the laptop:** local training runs at background priority, can pause and resume, and stops itself when it starts memorising instead of learning.
- **Downloads that survive bad internet:** resumable transfers that retry until every byte arrives.

05

A SECOND BRAIN FOR SHORT SIGNS

"Hello" finally works, and it answers in sentences

The sentence model learned from how-to videos, so it barely knew "hello". I put the 250-sign model into the app alongside it.

- **Word mode:** right first time for 61% of signs by people it had never seen, and in its top five 80% of the time. One real camera test read "dog" at 94%.
- **Single signs become full sentences:** "hello" offers "Hello!" or "Hi, how are you doing?", "thirsty" offers "Can I have some water?".
- **Smart ordering:** short signing shows sign suggestions first, long signing shows sentences first, and very unlikely guesses are hidden.

06

BUILT AROUND THE SIGNER

I rebuilt the experience so you never stop signing

- **Keep signing, review later:** translations pile up in one tidy, scrolling panel instead of blocking the screen. The newest is open; older ones fold away.
- **Accept one, or send all as one message,** with a preview of the combined sentence.
- **Choose with your hand:** hold up 1, 2 or 3 fingers to pick a suggestion. It only counts if your hand is held still, so normal signing never triggers it by accident.
- **See what Visibu sees:** thin coloured points on every finger (thumb, index, middle, ring, pinky) show exactly what is being tracked.
- **Cleaner, calmer layout** on desktop and phone, with every new option switchable in Accessibility settings.
- **Accessibility kept intact:** keyboard access, screen-reader announcements that never steal focus, high contrast, reduced motion and large touch targets.

07

THE HEADLINE FEATURE

My signs: teach Visibu your own language

Because a demo will get things wrong, the most important improvement is that people can correct it and teach it.

1 · Type
What the sign means, like “Can I get a latte, please?”

2 · Show it
Sign it to the camera three times.

3 · Use it
Sign it in Translate and your meaning comes up first, as “Your sign”.

> **“This sign already has a meaning. Do you want to change it?”**
>
> If a new sign looks like one you already taught, or one Visibu already knows, it asks before changing anything: change it, keep both, or cancel.

- **Fingerprints, not video:** each sign is saved as 256 numbers describing the movement, in your own browser.
- **Measured, not guessed:** the “same sign” threshold lets through only 1% of different signs in testing, while 99% of repeats still matched (on simulated repeats; real ones vary more).
- **Full control:** rename, re-record, delete, or try your signs to see if they are recognised. A shaky take is flagged so you can redo it.

THE SCOREBOARD

Real numbers, measured on people and sentences the models never saw

61%
single signs right on the first guess

80%
right sign within the top five

+72%
sentence score after full How2Sign training

30/30
browser and training versions gave identical suggestions

<1 s
to load both models in the app

65
automated tests passing (54 app, 11 training)

SENTENCE MODEL, 1,713 UNSEEN SENTENCES	BEFORE	AFTER
Training sentences	4,418	**29,547**
BLEU (overall translation score)	1.85	**3.18**
chrF (character overlap)	16.2	**20.4**
ROUGE-L (word order overlap)	13.0	**15.7**
Vocabulary	3,459 words	**7,608 words**

Real example after the upgrade: “My name is Sylvia Russell and this is how you can use temporary hair extensions.” became “Hi, my name is sylvia russell and this is how you can use a hair color.” Most full sentences are still wrong; published research systems on this dataset score about 8 to 15 BLEU.

08

BUILT PROPERLY

Private, tested and documented

- **Privacy:** no video leaves the device, there is no cloud inference, models are self-hosted, and nothing is ever spoken or shown to the partner until the signer accepts it.
- **Licences tracked:** every dataset is credited with its terms. How2Sign is non-commercial, and that is flagged for the competition.
- **Reproducible:** the whole pipeline (download, import, train, evaluate, export, verify, install) is scripted and written up in the ML guide.

It still makes mistakes. Now it can learn from them.

Visibu is a research demo, not tested with Deaf signers yet, and it does not replace a qualified ASL interpreter. What changed is that it now understands everyday signs, offers real choices, keeps the signer in control, and grows with every sign someone teaches it. Next on the table: animating text back into signs.